

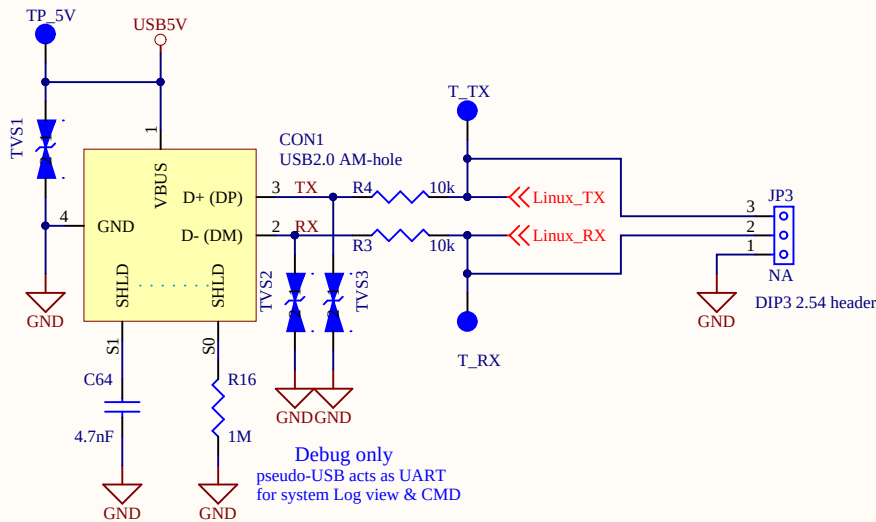
Block Diagram



Z_MB0.1
bare PCB

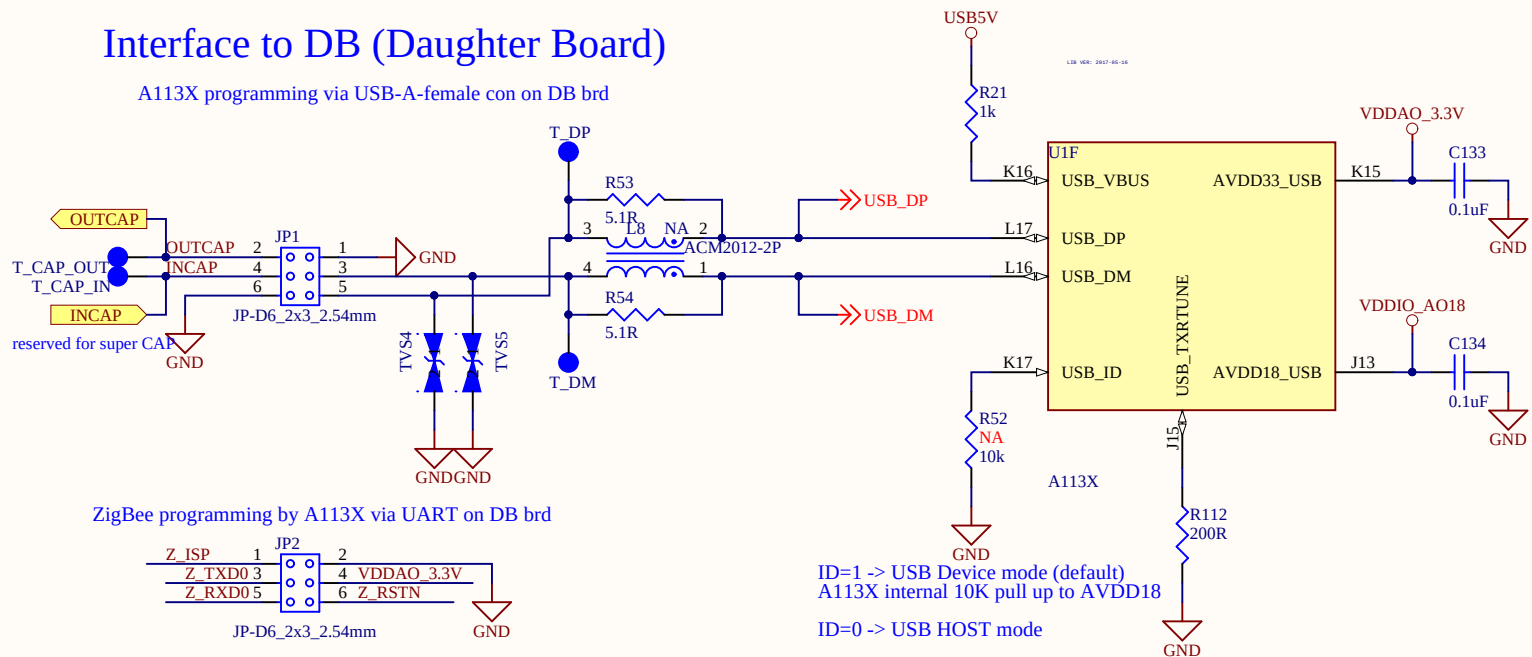
PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	BlockDiagram.SchDoc		
TITLE:	Block Diagram		
SIZE: A4	REVISION: 0.1	PAGE: 1 of 13	
DRAWN BY:	Zhou Tao	DATE:	2022/11/28

USB IN & OUT



Interface to DB (Daughter Board)

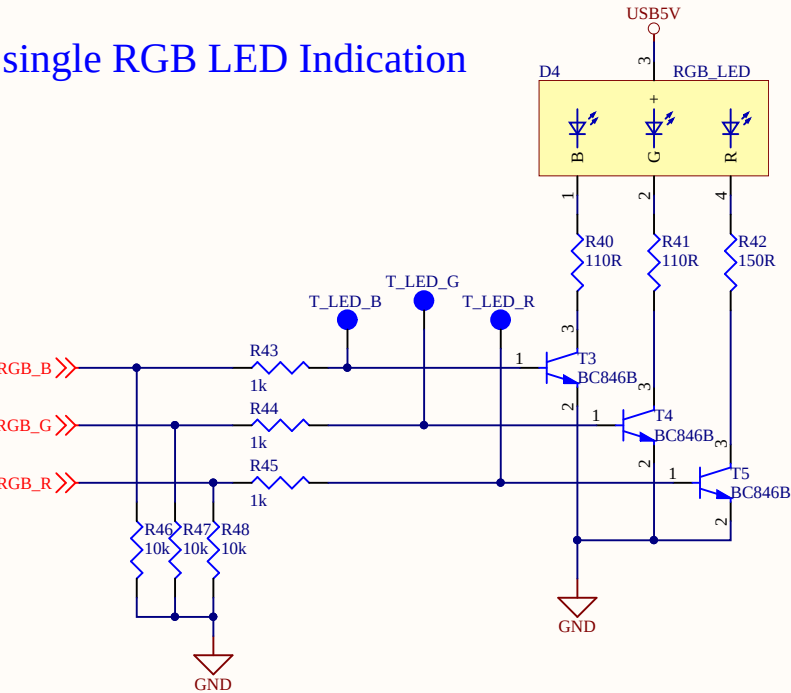
A113X programming via USB-A-female con on DB brd



ZigBee programming by A113X via UART on DB brd

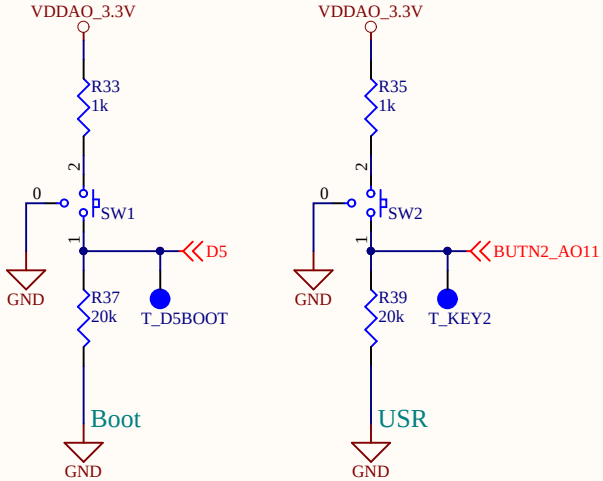
PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	USB_IN_OUT.SchDoc		
TITLE:	USB IN and OUT		
SIZE: A4	REVISION: 0.1	PAGE: 2 of 13	
DRAWN BY: Zhou Tao	DATE:		2022/11/28

single RGB LED Indication



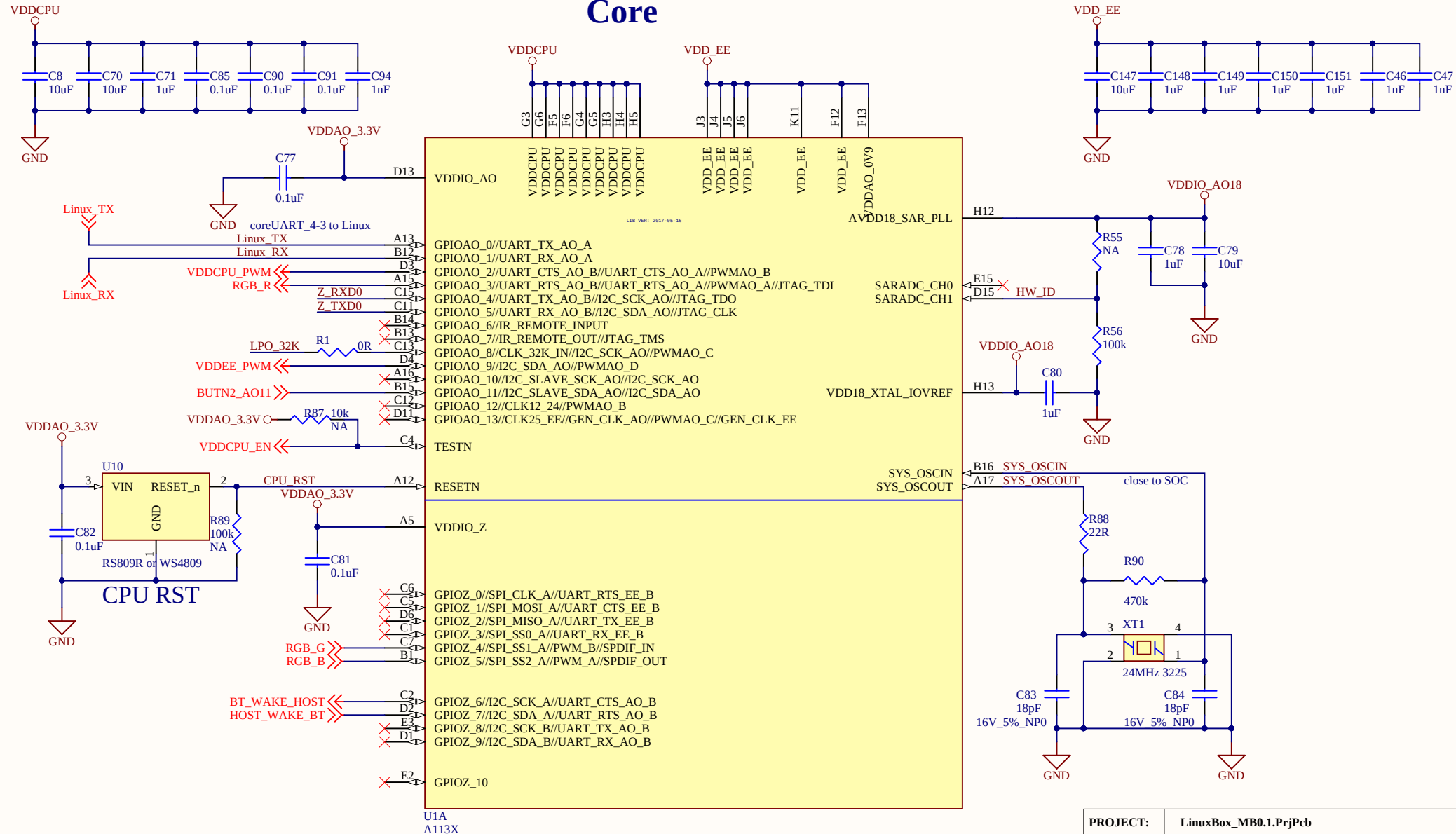
UI

dual KEYs for Unlock & USR Button



PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	UI.SchDoc		
TITLE:	UI		
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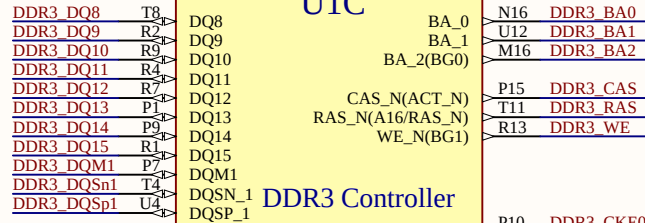
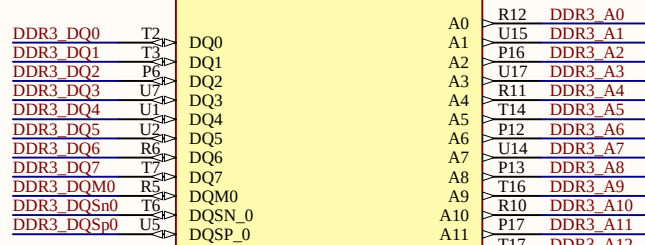
Core



PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	Core.SchDoc		
TITLE:	Core		
SIZE: A4	REVISION: 0.1	PAGE: 4 of 13	
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DDR3 - 1

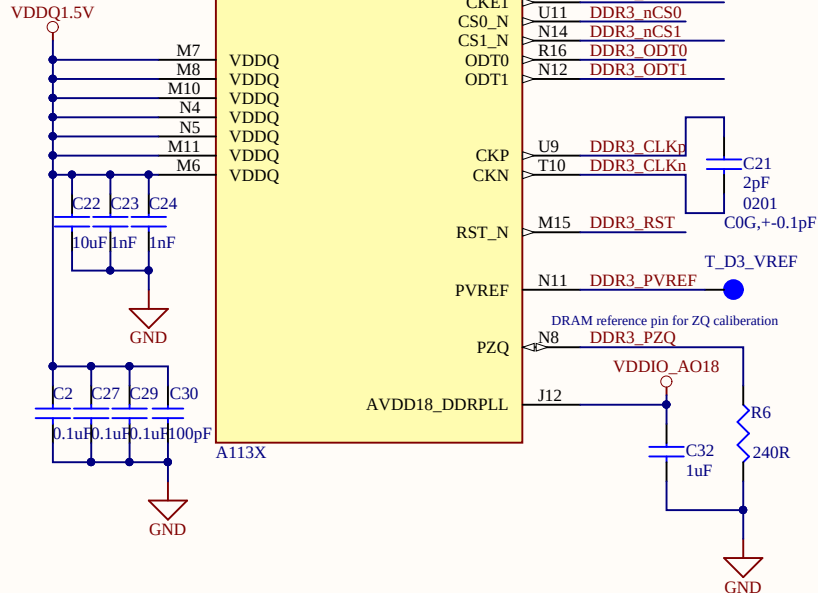
A



U1C

CAS_N(ACT_N)
RAS_N(A16/RAS_N)
WE_N(BG1)

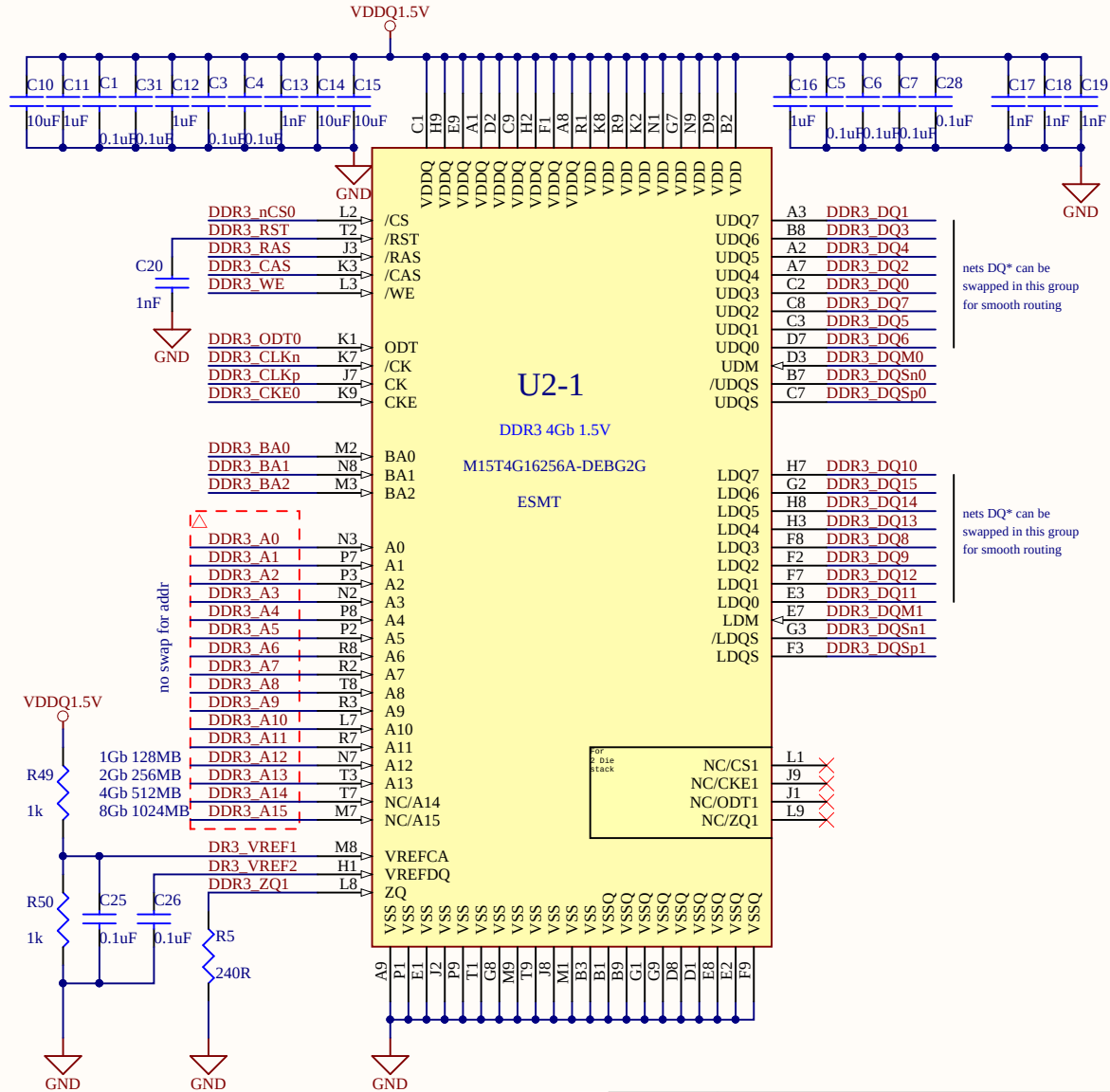
DDR3 Controller



2

3

4



U2-1

DDR3 4Gb 1.5V

M15T4G16256A-DEBG2G

ESMT

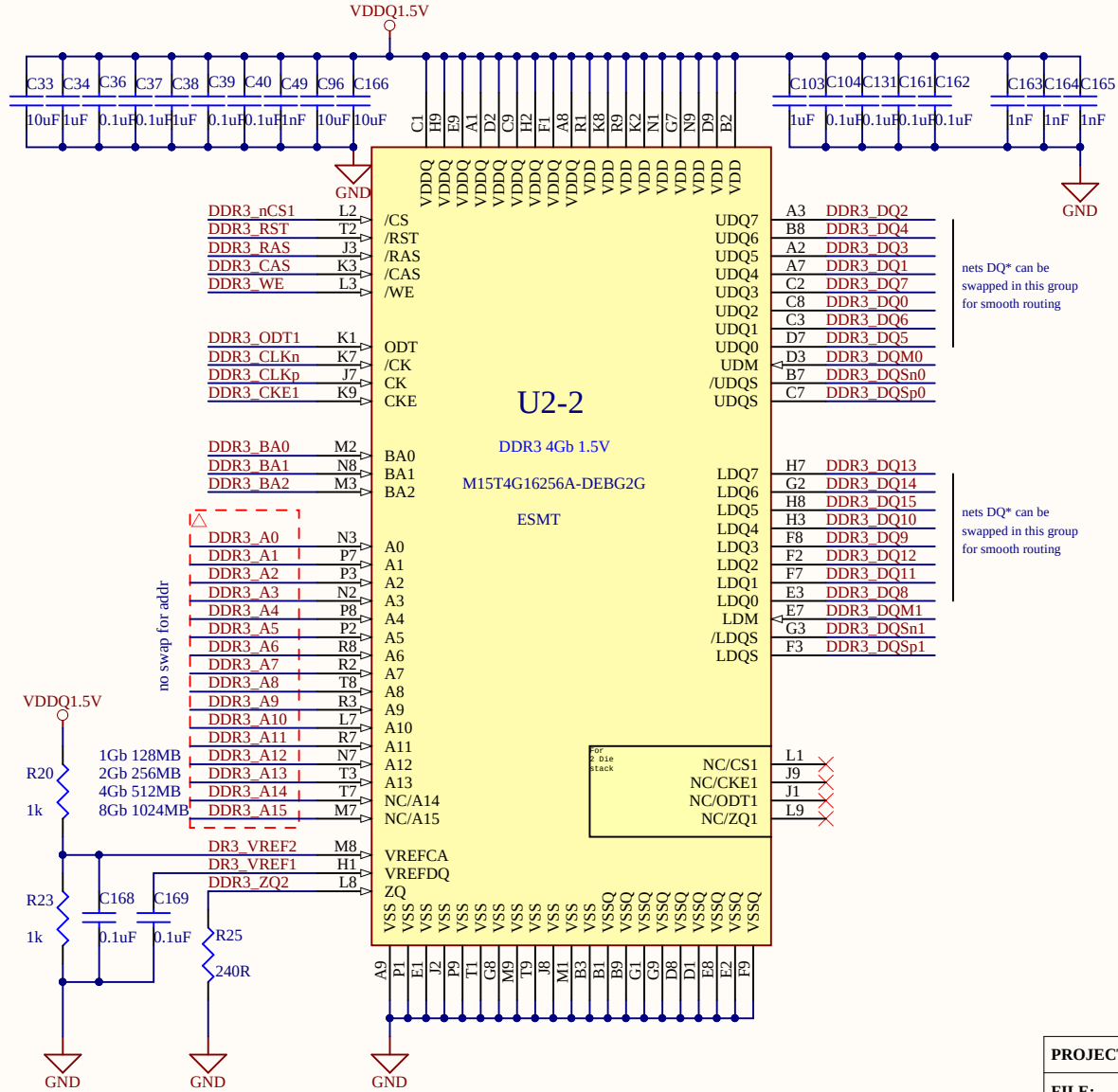
Basic layout rules:

1. CLK/DQS intra len mismatch < 0.5mm
2. CLK len > DQS len
3. Data lane len mismatch < 10mm, AC lane < 40mm

PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	DDR_1.SchDoc		
TITLE:	DDR3		
SIZE: A4	REVISION: 0.1	PAGE: 5 of 13	
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D

DDR3 - 2 (optional)



Basic layout rules:

1. CLK/DQS intra len mismatch < 0.5mm
2. CLK len > DQS len
3. Data lane len mismatch <10mm, AC lane < 40mm

PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	DDR_2.SchDoc		
TITLE:	DDR3		
SIZE: A4	REVISION: 0.1	PAGE: 6 of 13	
DRAWN BY:	Zhou Tao	DATE:	2022/11/28

A

A

B

B

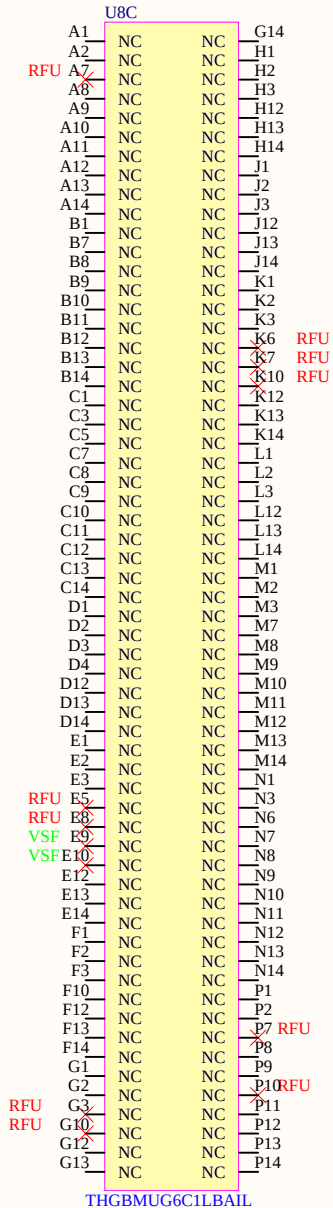
C

C

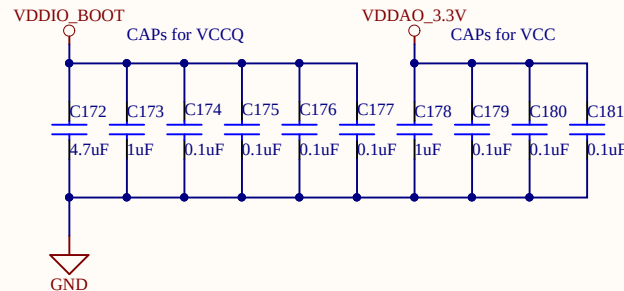
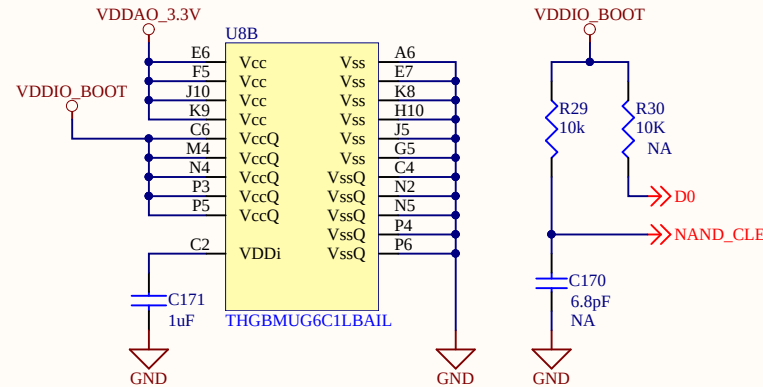
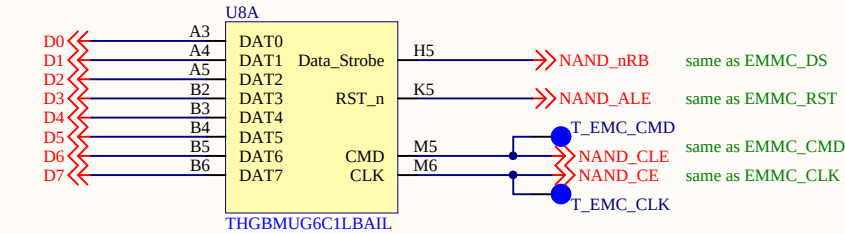
D

D

Flash (EMMC)



NC pins can be routed cross
but do NOT route over RFU & VSF pins

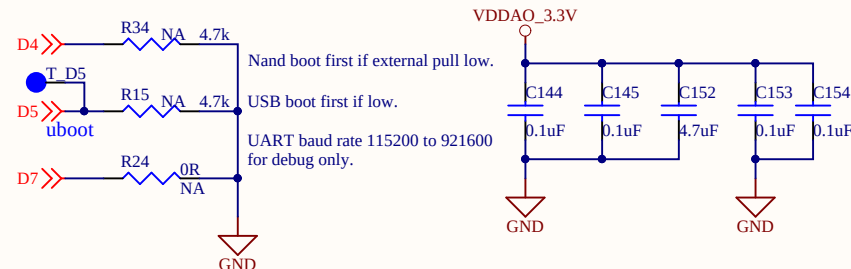
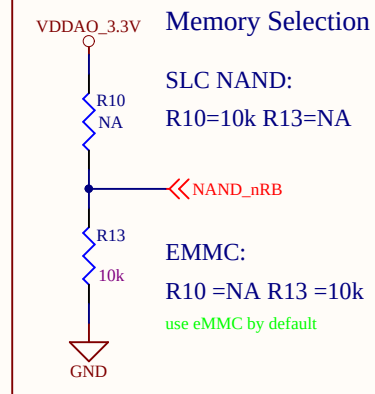
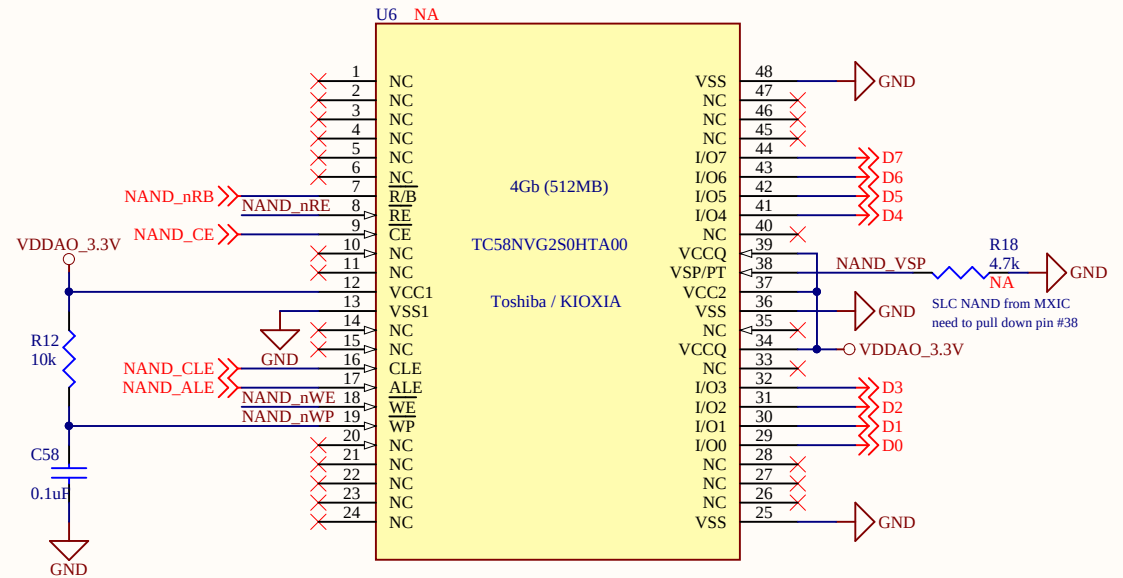
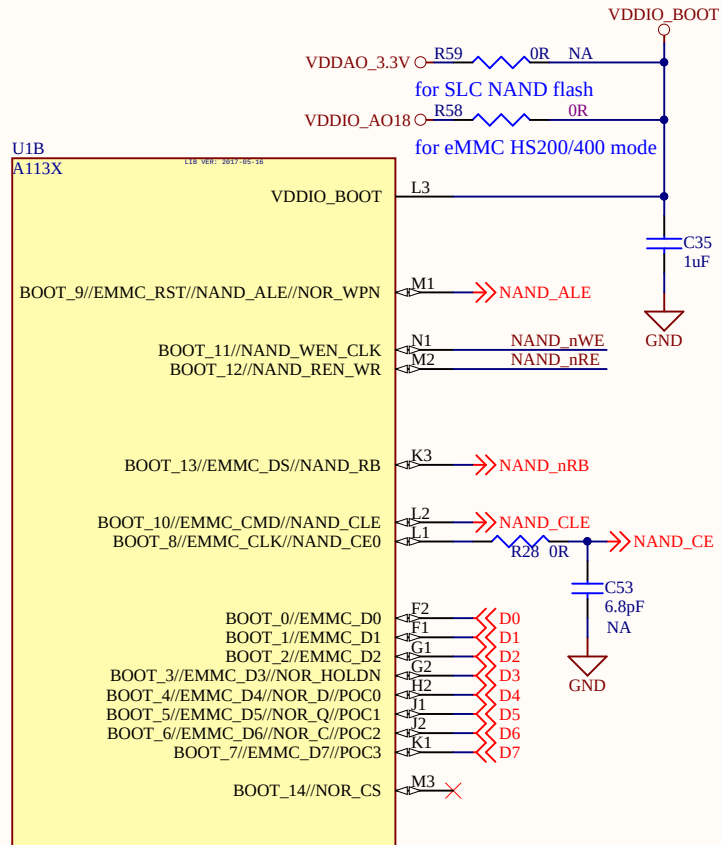


PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	Flash(EMMC).SchDoc		
TITLE:	Flash(EMMC)		
SIZE: A4	REVISION: 0.1	PAGE: 7	of 13
DRAWN BY: Zhou Tao	DATE:		2022/11/28

Flash (SLC NAND)

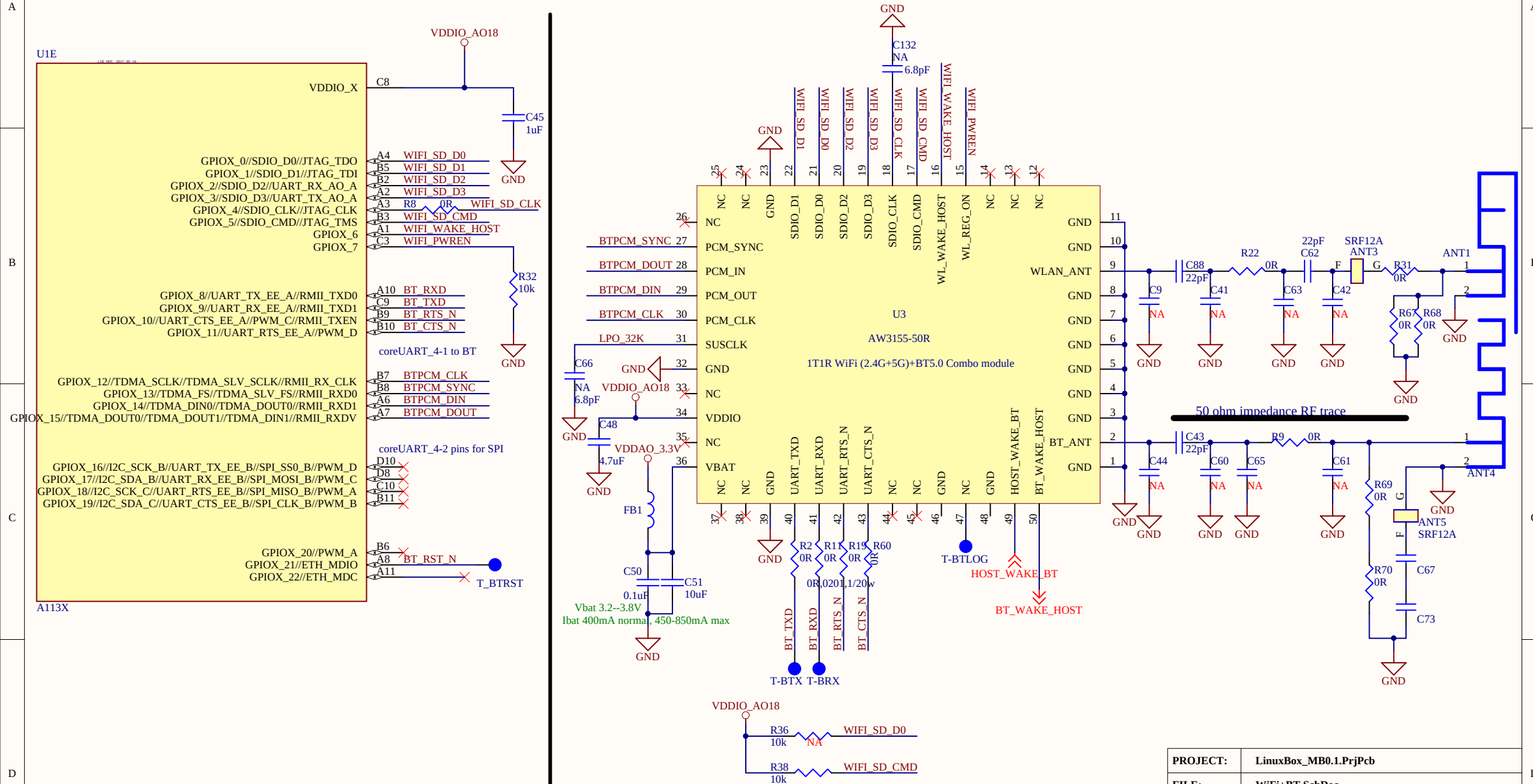
MLC NAND not supported

A113X only support SLC NAND flash

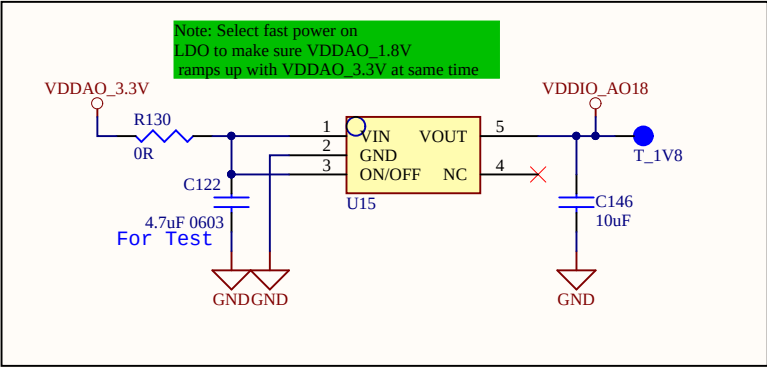
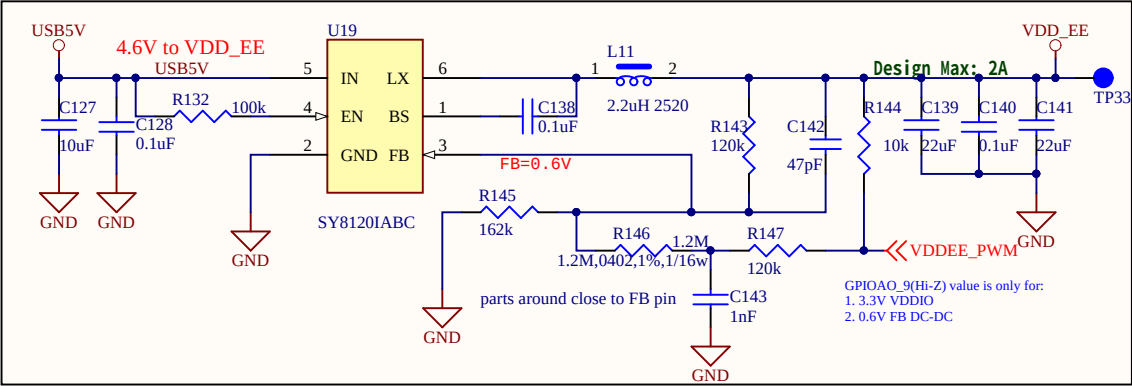
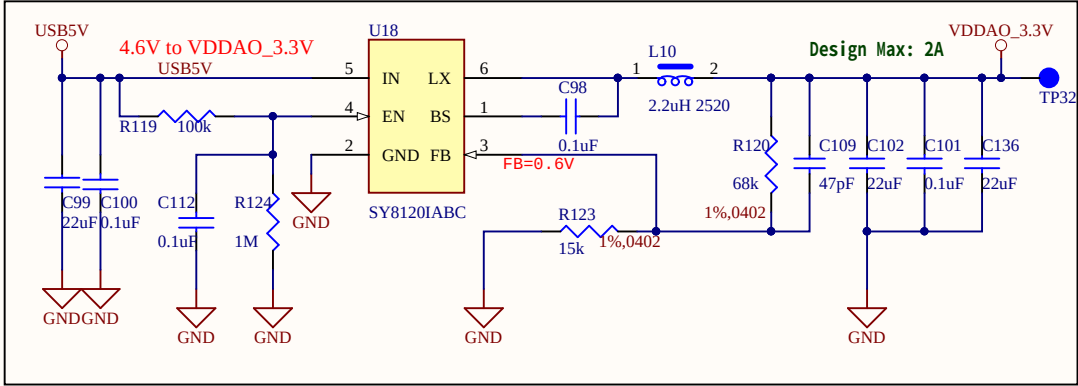
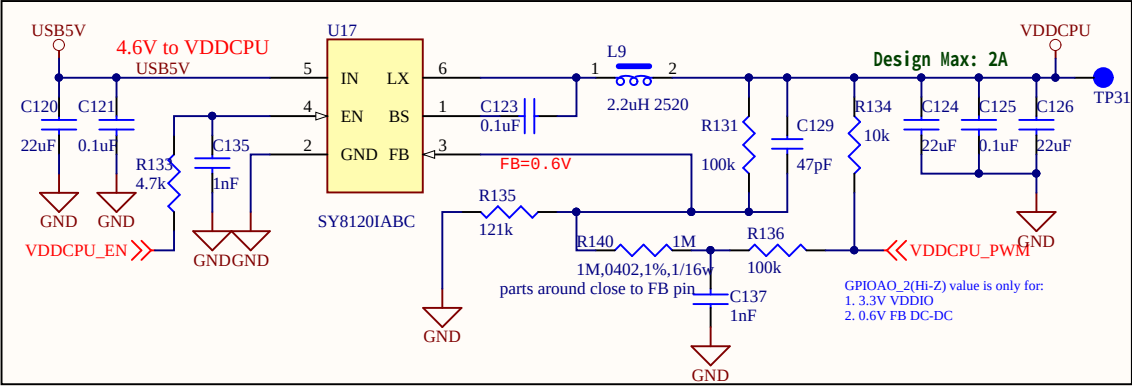
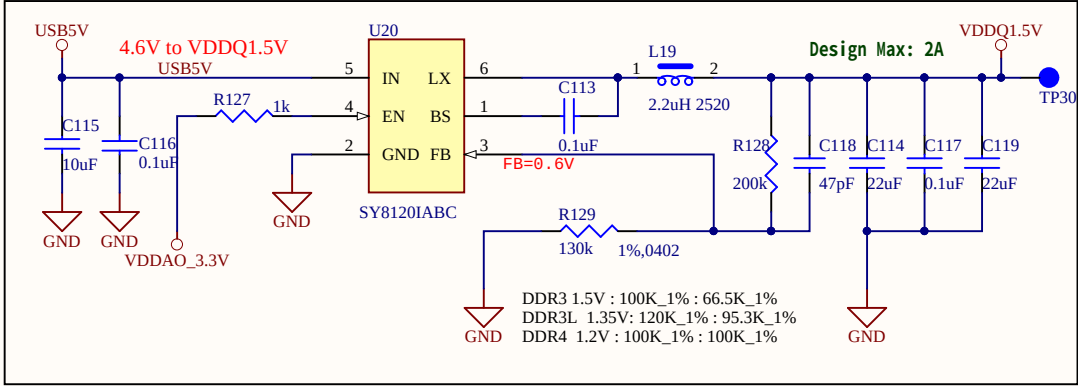


PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	Flash(NAND).SchDoc		
TITLE:	Flash(NAND)		
SIZE: A4	REVISION: 0.1	PAGE: 8	of 13
DRAWN BY: Zhou Tao	DATE:		2022/11/28

WiFi+BT

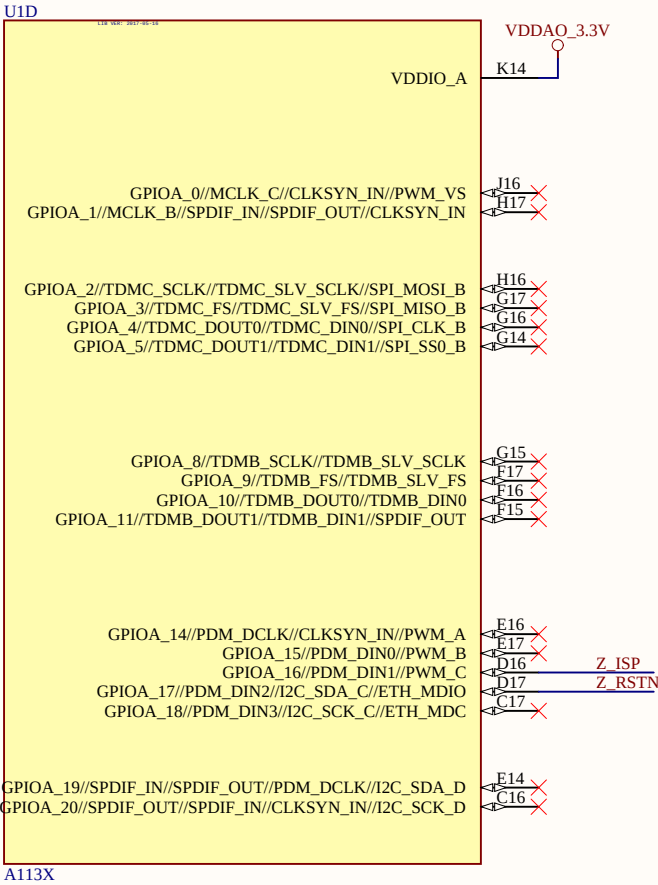


PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	WiFi+BT.SchDoc		
TITLE:	WiFi + BT		
SIZE: A4	REVISION: 0.1	PAGE: 9	of 13
DRAWN BY: Zhou Tao	DATE:		2022/11/28

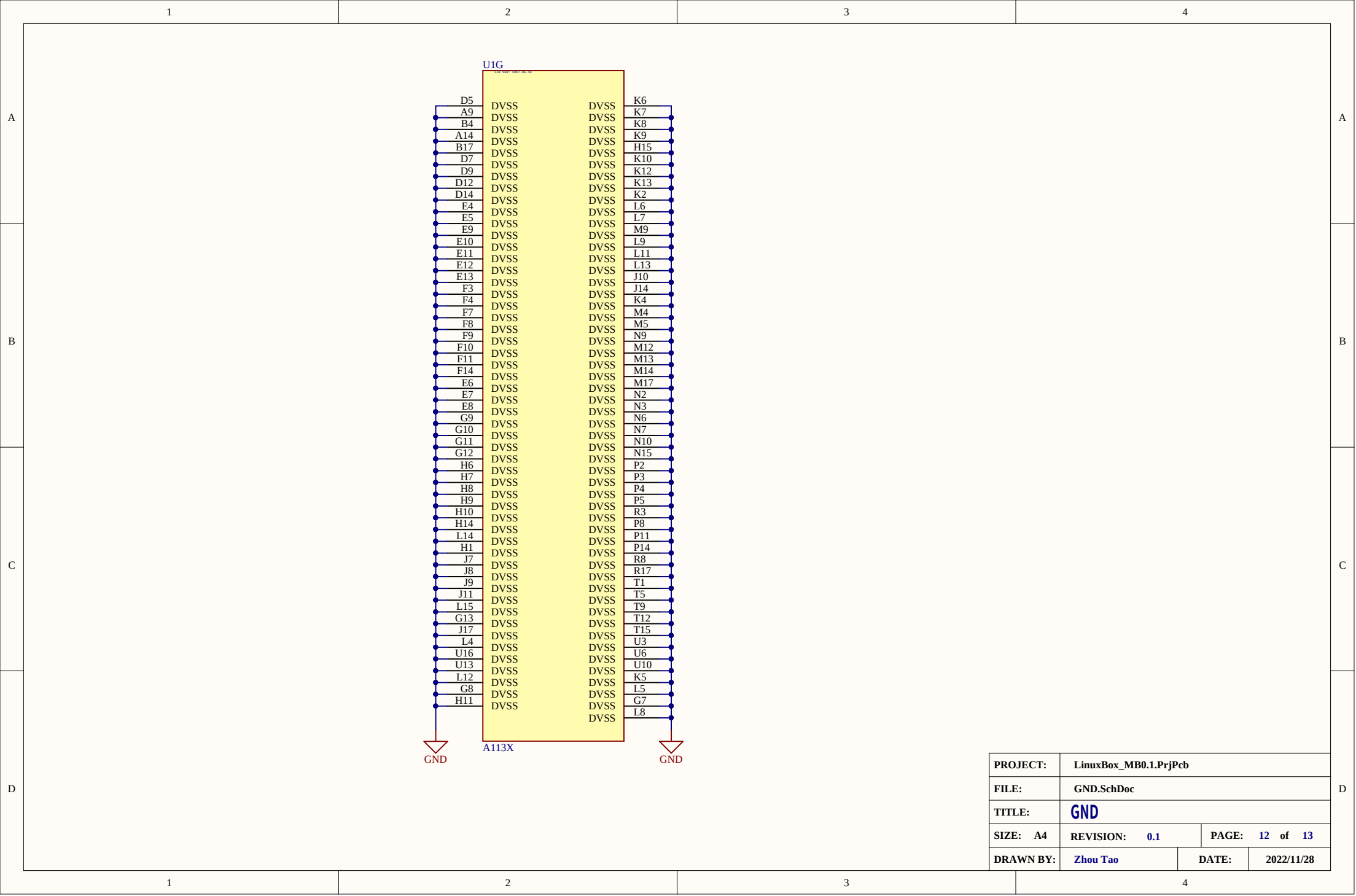


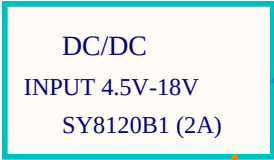
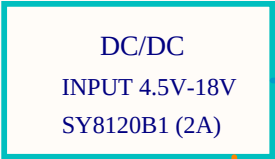
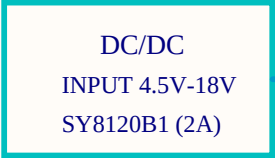
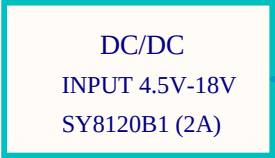
PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	Power.SchDoc		
TITLE:	Power		
SIZE: A4	REVISION: 0.1	PAGE: 10 of 13	
DRAWN BY: Zhou Tao	DATE:		2022/11/28

Audio MIC



PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	Audio.SchDoc		
TITLE:	Audio		
SIZE: A4	REVISION: 0.1	PAGE: 11 of 13	
DRAWN BY:	Zhou Tao	DATE:	2022/11/28





VDDAO_3.3V

VDD_EE

VDD_EE pin @500mA

VDDAO_0V9 pin @20mA

VDDQ1.5V

Max: 600mA

VDDCPU

1600mA

VDDAO_3.3V

TESTN VDDCPU_EN

VDDIO_AO pin @10mA
VDDIO_X pin(Optional)
WIFI_VBAT SDIO AC WiFi 800mA

VDDIO_A pin
UI

LDO
INPUT 2.5V-5.5V
WL2803E18-5 (500mA)

VDDIO_AO18

VDDIO_X pin @12mA
VDD18_XTAL_IOVREF pin @10mA
AVDD18_DDRPLL pin @30mA
AVDD18_USB pin @20mA
AVDD18_SAR_PLL pin @40mA
WIFI_VDDIO

VDDAO_3.3V

VDDIO_Z

AVDD33_USB pin
VDDIO_BOOT pin
VDDIO_Y pin(Optional)
NAND VCC_PWR
Audio VCC_3.3V_PWR
DAC: VCC_3.3V_PWR

PROJECT:	LinuxBox_MB0.1.PrjPcb		
FILE:	Power Tree.SchDoc		
TITLE:	Power Tree		
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DRAWN BY:	Zhou Tao	DATE:	2022/11/28

